

Cloud Native Spring

Crafting the Perfect Java Image for K8s

bellsoft

whoami



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BellSoft

- Vendor of Liberica JDK
- Contributor to the OpenJDK
- Author of ARM32 support in JDK
- Own base images
- Own Linux: Alpaquita

Liberica is the JDK officially recommended
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We know our stuff!



So, what is ultimate?

- Smallest
- Fatsest startup
- Something inbetween

So, let's look at all of
them!

How do we create an image?

Let's start from trivial.

```
1 FROM bellsoft/liberica-runtime-container:jdk-26-musl
2
3 COPY . /app
4 RUN cd /app && ./gradlew build
5 ENTRYPOINT java -jar /app/build/libs/spring-petclinic*.jar
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```
1 FROM bash
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3 COPY . /app
4 RUN rm -rf /app
```

4. We would like image to be light, but it's not :(

Our Dockerfile

```
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```


Result

1	Cmp	Size	Command
2		8.6 MB	FROM blobs
3		112 MB	RUN /bin/sh -c cd /app && ./gradlew build # buildkit
4		88 MB	[2/3] COPY spring-petclinic /app
5		580 MB	mount / from exec /bin/sh -c cd /app && ./gradlew build

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112 MB of Java

88 MB of the app

580 MB of Gradle build caches

600+ MB are changed on every build!

Why do we care? We care because:

1. `push` takes a longer time to start (update is longer)
2. `pull` takes longer (update takes longer & scaling takes longer)

Also, more disk space is inefficiently used



Optimizing. Round 1.

Let's build it outside of container

```
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3 COPY . /app
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```

Optimizing. Round 1.

Let's build it outside of container

```
1 FROM bellsoft/liberica-runtime-container:jdk-26-musl
2
3 COPY build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
4 CMD java -jar /app/app.jar
```

Optimizing. Round 1.

Let's build it outside of container

```
1 FROM bellsoft/liberica-runtime-container:jdk-26-musl
2
3 COPY build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
4 CMD java -jar /app/app.jar
```

Just copying the prebuilt `jar` file

Results

1	Cmp	Size	Command
2		10 MB	FROM blobs
3		109 MB	cd /app && ./gradlew build -xtest
4		85 MB	(missing)
5		591 MB	(missing)

No Gradle caches and build dir!

Results

1	Cmp	Size	Command
2		8.6 MB	FROM blobs
3		112 MB	pulled from docker.io/bellsoft/liberica-runtime-container:jdk-26-musl@sh
4		70 MB	[2/2] COPY spring-petclinic/build/libs/spring-petclinic-4.0.0-SNAPSHOT.j

No Gradle caches and build dir!

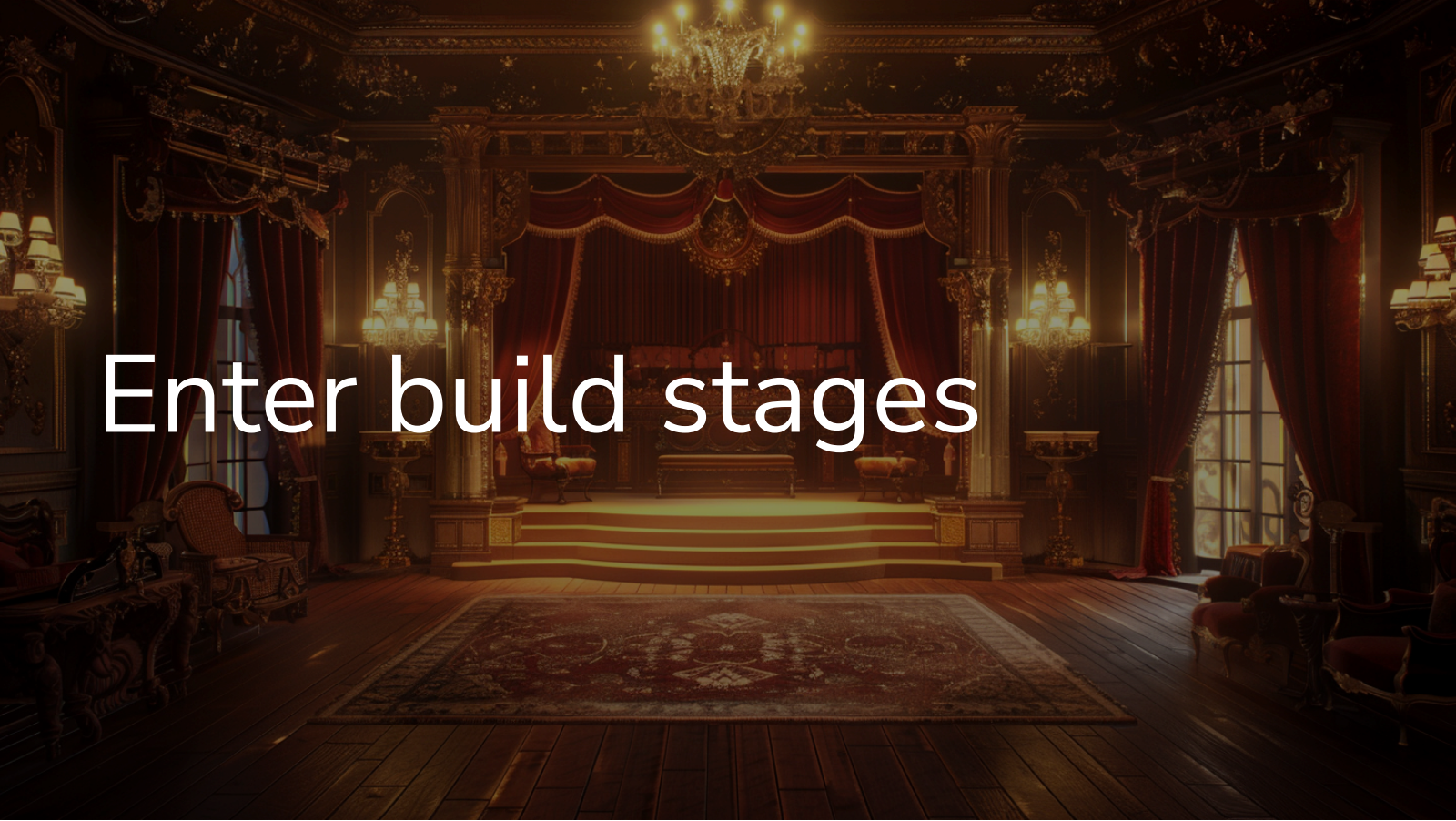
Saved 600+ MB

But the build is not clean now :(

Saved 600+ MB

But the build is not clean now :(

We have to build everything outside, what if environment affects the build?

A highly detailed, ornate theater stage set. The stage is framed by a grand, dark wood structure with classical columns and intricate carvings. Red velvet curtains hang from the top, framing the stage area. A large, multi-tiered chandelier hangs from the ceiling, casting a warm, golden light. The stage floor is made of dark wood, and a large, patterned rug is laid out in the foreground. The overall atmosphere is one of grandeur and elegance.

Enter build stages

Multi-stage builds

Holy grail of pure builds

- Allow clean builds
- Allow optimal packaging
- Allow different base images

Staged example

```
1 FROM bellsoft/liberica-runtime-container:jdk-26-musl as builder
2
3 COPY . /app
4 RUN cd /app && ./gradlew build -xtest
5
6 FROM bellsoft/liberica-runtime-container:jre-26-musl as runner
7
8 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
```

Staged example

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1 FROM bellsoft/liberica-runtime-container:jdk-26-musl as builder
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8 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
```

Result

- jar - same size
- JRE - 48.28

That's all folks!

That's all folks!

Or is it?

What's these 60 MiB?

We have to pull 60 MiB of petclinic on every small commit!

Is there a way to optimize it?

Layers!

```
1 FROM bellsoft/liberica-runtime-container:jdk-26-musl as builder
2
3 COPY . /app
4 RUN cd /app && ./gradlew build -xtest
5
6 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as optimizer
7
8 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
9 WORKDIR /app
10 RUN java -Diarmode=tools -jar /app/app.jar extract --layers --destination extracted
```

- Build image

Layers!

```
1 FROM bellsoft/liberica-runtime-container:jre-26-musl as builder
2
3 COPY . /app
4 RUN cd /app && ./gradlew build -xtest
5
6 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as optimizer
7
8 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
9 WORKDIR /app
10 RUN java -Djarmode=tools -jar /app/app.jar extract --layers --destination extracted
11
```

- Build image
- Introduce new "optimizer" stage

Layers!

```
4  RUN cd /app && ./gradlew build -x test
5
6  FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as optimizer
7
8  COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
9  WORKDIR /app
10 RUN java -Djarmode=tools -jar /app/app.jar extract --layers --destination extracted
11
12 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as runner
13
14 ENTRYPOINT ["java", "-jar", "/app/spring-petclinic.jar"]
```

- Build image
- Introduce new "optimizer" stage

Layers!

```
5
6 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as optimizer
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8 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
9 WORKDIR /app
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11
12 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as runner
13
14 ENTRYPOINT ["java", "-jar", "./app/spring-petclinic.jar"]
15 COPY --from=optimizer /app/extracted/dependencies/ /
```

- Build image
- Introduce new "optimizer" stage
- Extract the jar to layered structure

Layers!

```
8 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
9 WORKDIR /app
10 RUN java -Djarmode=tools -jar /app/app.jar extract --layers --destination extracted
11
12 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as runner
13
14 ENTRYPOINT ["java", "-jar", "./app/spring-petclinic.jar"]
15 COPY --from=optimizer /app/extracted/dependencies/ ./
16 COPY --from=optimizer /app/extracted/spring-boot-loader/ ./
17 COPY --from=optimizer /app/extracted/snapshot-dependencies/ ./
```

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- Introduce new "optimizer" stage
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Layers!

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18 COPY --from=optimizer /app/extracted/application/ ./
```

- Build image
- Introduce new "optimizer" stage
- Extract the jar to layered structure
- Copy layers

Layers!

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12 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as runner
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14 ENTRYPOINT ["java", "-jar", "./app/spring-petclinic.jar"]
15 COPY --from=optimizer /app/extracted/dependencies/ ./
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18 COPY --from=optimizer /app/extracted/application/ ./
```

- Build image
- Introduce new "optimizer" stage
- Extract the jar to layered structure
- Copy layers

Layers

Because Spring Boot jar is complex!

```
1  Usage:
2    java -Djarmode=tools -jar my-app.jar
3
4  Available commands:
5    extract      Extract the contents from the jar
6    list-layers  List layers from the jar that can be extracted
```


Layers

```
1  extracted
2  |— application
3    |— spring-petclinic-4.0.0-SNAPSHOT.jar
4  |— dependencies
5    |— lib
6        |— angus-activation-2.0.2.jar
7        |— antlr4-runtime-4.13.2.jar
8        |— aspectjweaver-1.9.24.jar
9        |— attoparser-2.0.7.RELEASE.jar
10       |— bootstrap-5.3.8.jar
11       |— byte-buddy-1.17.7.jar
12       |— cache-api-1.1.1.jar
13       |— caffeine-3.2.2.jar
14       |— checker-qual-3.49.3.jar
15       |— classmate-1.7.0.jar
16       |— commons-logging-1.3.5.jar
```

Layers

```
91  — spring-orm-7.0.1.jar
92  — spring-tx-7.0.1.jar
93  — spring-web-7.0.1.jar
94  — spring-webmvc-7.0.1.jar
95  — thymeleaf-3.1.3.RELEASE.jar
96  — thymeleaf-spring6-3.1.3.RELEASE.jar
97  — tomcat-embed-core-11.0.14.jar
98  — tomcat-embed-el-11.0.14.jar
99  — tomcat-embed-websocket-11.0.14.jar
100 — txw2-4.0.5.jar
101 — unescape-1.1.6.RELEASE.jar
102 — webjars-locator-lite-1.1.1.jar
103 — snapshot-dependencies
104 — spring-boot-loader
105
106 6 directories, 98 files
```

Layers

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91      — spring-orm-7.0.1.jar
92      — spring-tx-7.0.1.jar
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Layers

```
3  spring-petclinic-4.0.0-SNAPSHOT.jar
4  └─ dependencies
5      └─ lib
6          └─ angus-activation-2.0.2.jar
7          └─ antlr4-runtime-4.13.2.jar
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15         └─ classmate-1.7.0.jar
16         └─ commons-logging-1.3.5.jar
17         └─ context-propagation-1.2.0.jar
18         └─ error_prone_annotations-2.40.0.jar
19         └─ font-awesome-4.7.0.jar
```

Layers

```
1  extracted
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```

Layered image structure

```
1  > du -h --max-depth=1 extracted
2  66M    extracted/dependencies
3  0      extracted/spring-boot-loader
4  0      extracted/snapshot-dependencies
5  892K   extracted/application
6  66M    extracted
```

Layered image structure

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```

- Dependencies: ~66MiB

Layered image structure

```
1  > du -h --max-depth=1 extracted
2  66M    extracted/dependencies
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5  892K   extracted/application
6  66M    extracted
```

- Dependencies: ~66MiB
- Application: ~892K

66.0MiB > 60Mib!



What are we optimizing?

Pull size!

Pull size here is usually only around 1MiB!

We just reinvented how the BellSoft's buildpack works!

And it is amazing

Diversion: buildpacks

<https://paketo.io/>

Trivial usage:

```
1  /usr/sbin/pack build petclinic \  
2    --builder bellsoft/buildpacks.builder:musl \  
3    --path . \  
4    -e BP_JVM_VERSION=26
```

Diversion: buildpacks

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Result

1	ID	TAG	SIZE	COMMAND
2	2774e3f214	petclinic:latest	0B	Buildpacks Process Types
3	<missing>		1.44kiB	Buildpacks Launcher Config
4	<missing>		2.44MiB	Buildpacks Application Launcher
5	<missing>		59.17MiB	Application Layer
6	<missing>		729.21kiB	Software Bill-of-Materials
7	<missing>		97.87MiB	Layer: 'jre', Created by buildpack: bellsoft/buildpack
8	<missing>		200.00B	Layer: 'java-security-properties', Created by buildpack
9	<missing>		4.27MiB	Layer: 'helper', Created by buildpack: bellsoft/buildpack
10	<missing>		2.97kiB	
11	<missing>		7.48MiB	

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5	<missing>		59.17MiB	Application Layer
6	<missing>		729.21kiB	Software Bill-of-Materials
7	<missing>		97.87MiB	Layer: 'jre', Created by buildpack: bellsoft/buildpack
8	<missing>		200.00B	Layer: 'java-security-properties', Created by buildpack
9	<missing>		4.27MiB	Layer: 'helper', Created by buildpack: bellsoft/buildpack
10	<missing>		2.97kiB	
11	<missing>		7.48MiB	

Buildpacks

Why do we need them?

- Buildpacks transform your application source code into container images
 - The Paketo open source project provides production-ready buildpacks for the most popular languages and frameworks
 - Use Paketo Buildpacks to easily build your apps and keep them updated
- Reminds of s2i images
 - Build better than default
 - Spring-aware

Is this all?

We've optimized pull/push size, right?

Optimization is multidimensional

When startup time is more important...

There are several solutions for the Spring application startup time

1. Class Data Sharing (CDS)
2. Project Leyden
3. Coordinated Restore at Checkpoint
4. Native Image

Class Data Sharing (CDS)

- When: Java 5 (!)
- Why: reduce the startup and memory footprint of multiple JVM instances running on the same host
- How: stores data in an archive

How does it help us?

Class Data Sharing (CDS)

- When: Java 5 (!)
- Why: reduce the startup and memory footprint of multiple JVM instances running on the same host
- How: stores data in an archive

How does it help us?

It does not! 🤖

Class Data Sharing (CDS)

- When: Java 5 (!)
- Why: reduce the startup and memory footprint of multiple JVM instances running on the same host
- How: stores data in an archive

How does it help us?

It does not! 🤖 But

Application Class Data Sharing

- When: Java 10
- Why: add application classes to the archive

And then:

- When: Java 13, JEP 350
- Why: allow addition of classes into the archive upon app exit!

And this helps! How?

How to use AppCDS with Spring?

- `-XX:ArchiveClassesAtExit=application.jsa` to create an archive
- `-Dspring.context.exit=onRefresh` to start and immediately exit the application

NB:

1. Use the same JDK
2. Use the same arguments

And even better!

Spring AOT

Add `-Dspring.aot.enabled=true`

Even more classes!!!

Practice

```
1 FROM bellsoft/liberica-runtime-container:jdk-26-musl as builder
2
3 COPY . /app
4 RUN cd /app && ./gradlew build -xtest
5
6 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as optimizer
7
8 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
9 WORKDIR /app
10 RUN java -Djarmode=tools -jar /app/app.jar extract --layers --launcher
11
12 FROM bellsoft/liberica-runtime-container:jre-26-slim-musl as runner
13
14 ENTRYPOINT ["java", "-jar", "./app/app.jar"]
15 COPY --from=optimizer /app/extracted/dependencies/ ./
16 COPY --from=optimizer /app/extracted/spring-boot-loader/ ./
17 COPY --from=optimizer /app/extracted/snapshot-dependencies/ ./
18 COPY --from=optimizer /app/extracted/application/ ./
```

Practice

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1 FROM bellsoft/liberica-runtime-container:jdk-26-musl as builder
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9 WORKDIR /app
10 RUN java -Djarmode=tools -jar /app/app.jar extract --layers --destination extracted
11 FROM bellsoft/liberica-runtime-container:jre-26-cds-slim-musl as runner
12
13 ENTRYPOINT ["java", "-jar", "./app/app.jar"]
14 COPY --from=optimizer /app/extracted/dependencies/ ./
15 COPY --from=optimizer /app/extracted/spring-boot-loader/ ./
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Practice

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1 #omitted
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8 COPY --from=optimizer /app/extracted/application/ ./
9 RUN java -Dspring.aot.enabled=true \
10     -XX:ArchiveClassesAtExit=./application.jsa \
11     -Dspring.context.exit=onRefresh \
12     -jar app.jar
```

Practice

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2 FROM bellsoft/liberica-runtime-container:jre-26-cds-slim-musl as runner
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Practice

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7 COPY --from=optimizer /app/extracted/snapshot-dependencies/ ./
8 COPY --from=optimizer /app/extracted/application/ ./
9 RUN java -Dspring.aot.enabled=true \
10     -XX:ArchiveClassesAtExit=./application.jsa \
11     -Dspring.context.exit=onRefresh \
12     -jar app.jar
```

Practice

```
1 #omitted
2 FROM bellsoft/liberica-runtime-container:jre-26-cds-slim-musl as runner
3
4 ENTRYPOINT ["java", \
5             "-Dspring.aot.enabled=true", \
6             "-XX:SharedArchiveFile=application.jsa", \
7             "-jar", "app.jar"]
8 COPY --from=optimizer /app/extracted/dependencies / ./
9 COPY --from=optimizer /app/extracted/spring-boot-loader/ ./
10 COPY --from=optimizer /app/extracted/snapshot-dependencies/ ./
11 COPY --from=optimizer /app/extracted/application/ ./
12 RUN java -Dspring.aot.enabled=true \
13     -XX:ArchiveClassesAtExit=./application.jsa \
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Practice

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13     -XX:ArchiveClassesAtExit=./application.jsa \
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Practice

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Practice

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13     -XX:AOTCacheOutput=app.aot \
14     -Dspring.context.exit=onRefresh \
15     -jar app.jar
```

What does it cost ?

1 111M Nov 4 23:47 app.aot

Which is not small at all!

What does it cost ?

1 111M Nov 4 23:47 app.aot

Which is not small at all!

But you're trading pull speed for startup speed

Average startup time per variant

8

6

4

2

0

7.13

5.77

4.79

3.67

6.07

4.86

2.70

1.48

noaot

springaot

aotcache

aotcache+springaot

extracted noaot

extracted springaot

extracted aotcache

extracted
aotcache+springaot



Pushing further with CRaC

CRaC: Coordinated Restore at Checkpoint

The CRaC (Coordinated Restore at Checkpoint) Project researches coordination of Java programs with mechanisms to checkpoint (make an image of, snapshot) a Java instance while it is executing. Restoring from the image could be a solution to some of the problems with the start-up and warm-up times. The primary aim of the Project is to develop a new standard mechanism-agnostic API to notify Java programs about the checkpoint and restore events. Other research activities will include, but will not be limited to, integration with existing checkpoint/restore mechanisms and development of new ones, changes to JVM and JDK to make images smaller and ensure they are correct.

<https://openjdk.org/projects/crac/>

In a perfect world it should be:

```
1  FROM bellsoft/liberica-runtime-container:jdk-26-musl as builder
2
3  COPY . /app
4  RUN cd /app && ./gradlew build
5
6  FROM bellsoft/liberica-runtime-container:jre-26-crac-slim as optimizer
7
8  COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
9  WORKDIR /app
10 RUN java -Dspring.context.checkpoint=onRefresh -XX:CRaCCheckpointTo=/checkpoint -jar /app/app.jar
11
12 FROM bellsoft/liberica-runtime-container:jre-26-crac-slim as runner
13
14 ENTRYPOINT java -XX:CRaCRestoreFrom=/checkpoint
15 COPY --from=optimizer /app/app.jar /app/app.jar
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```

In a perfect world it should be:

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7
8  COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
9  WORKDIR /app
10 RUN java -Dspring.context.checkpoint=onRefresh -XX:CRaCCheckpointTo=/checkpoint -jar /app/app.jar
11
12 FROM bellsoft/liberica-runtime-container:jre-26-crac-slim as runner
13
14 ENTRYPOINT java -XX:CRaCRestoreFrom=/checkpoint
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16 COPY --from=optimizer /checkpoint /checkpoint
```

But in reality

```
1  #12 6.051      Suppressed: java.lang.RuntimeException: Native checkpoint failed.
2  #12 6.051      at java.base/jdk.crac.Core.translateJVMExceptions(Core.java:114) ~[r
3  #12 6.051      at java.base/jdk.crac.Core.checkpointRestore1(Core.java:192) ~[na:na
4  #12 6.051      at java.base/jdk.crac.Core.checkpointRestore(Core.java:299) ~[na:na
5  #12 6.051      at java.base/jdk.crac.Core.checkpointRestore(Core.java:278) ~[na:na
6  #12 6.051      at java.base/javax.crac.Core.checkpointRestore(Core.java:73) ~[na:na
7  #12 6.051      at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(
8  #12 6.051      at java.base/java.lang.reflect.Method.invoke(Method.java:580) ~[na:r
9  #12 6.051      at org.crac.Core$Compat.checkpointRestore(Core.java:141) ~[crac-1.4.
10 #12 6.051      ... 17 common frames omitted
```

CRaC is hard!

Let's try to fix it with arcane magic

```
1 FROM bellsoft/liberica-runtime-container:jdk-26-musl as builder
2
3 COPY . /app
4 RUN cd /app && ./gradlew build -xtest
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6 FROM bellsoft/liberica-runtime-container:jre-26-crac-slim as optimizer
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8 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
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CRaC is hard!

Let's try to fix it with arcane magic

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10 WORKDIR /app
11 RUN java -Dspring.context.checkpoint=onRefresh -XX:CRaCCheckpointTo=/checkpoint -jar /app/app.jar
12
13 FROM bellsoft/liberica-runtime-container:jre-26-crac-slim as runner
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14
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16 COPY --from=optimizer /app/app.jar /app/app.jar
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```


CRaC is hard!

Let's try to fix it with arcane magic

```
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2 FROM bellsoft/liberica-runtime-container:jdk-26-musl as builder
3
4 COPY . /app
5 RUN cd /app && ./gradlew build -xtest
6
7 FROM bellsoft/liberica-runtime-container:jre-26-crac-slim as optimizer
8
9 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
10 WORKDIR /app
11 RUN --security=insecure java -Dspring.context.checkpoint=onRefresh \
12     -XX:CRaCCheckpointTo=/checkpoint -jar /app/app.jar
13
14 FROM bellsoft/liberica-runtime-container:jre-26-crac-slim as runner
15
16 ENTRYPOINT java -XX:CRaCRestoreFrom=/checkpoint
```

CRaC is hard!

Let's try to fix it with arcane magic

```
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8
9 COPY --from=builder /app/build/libs/spring-petclinic-4.0.0-SNAPSHOT.jar /app/app.jar
10 WORKDIR /app
11 RUN --security=insecure java -Dspring.context.checkpoint=onRefresh \
12     -XX:CRaCCheckpointTo=/checkpoint -jar /app/app.jar || true
13
14 FROM bellsoft/liberica-runtime-container:jre-26-crac-slim as runner
15
16 ENTRYPOINT java -XX:CRaCRestoreFrom=/checkpoint
```

And this is not all!

Did you hear about `buildx` ?

```
1 docker buildx create --buildkitd-flags \  
2   '--allow-insecure-entitlement security.insecure' \  
3   --name insecure-builder
```

And this is not all!

Did you hear about `buildx` ?

```
1 docker buildx create --buildkitd-flags \  
2   '--allow-insecure-entitlement security.insecure' \  
3   --name insecure-builder
```

And this is not all!

Did you hear about `buildx` ?

```
1 docker buildx use insecure-builder
```

And this is not all!

Did you hear about `buildx` ?

```
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4   -t pet-crac --output type=docker .
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And this is not all!

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1 docker run --rm -it --privileged pet-crac
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5. Commit the container like

```
1  docker commit container-id new-tag
```

Pros and Cons

Pros:

1. Does not require arcane magic
2. Works more predictably
3. Does not depend on unstable features
4. Does not require privileged containers in

Cons:

1. Organization-specific
2. Requires more steps

Native image

Native image

1. No JRE dependency
2. Size roughly JRE+app (but can upx)
3. Can run on barebones Alpaquita
4. Long time to build

And now we have all
flavours of ultimate
docker images!

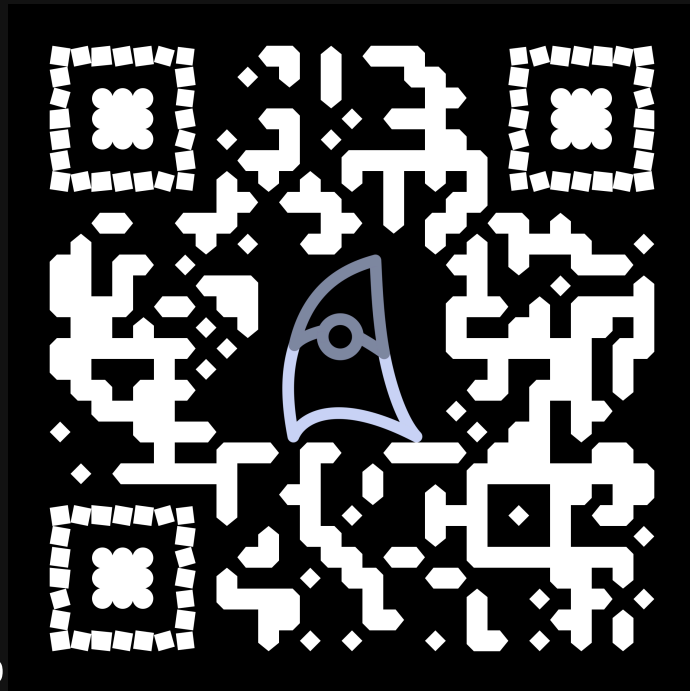
Quick summary?

1. Use layers for faster deployment
2. Use CDS for faster startup without many compromises
3. Use CRaC for a *lightning-fast* startup (with compromises)
4. Use native image if you're prepared to sacrifice some build time

Thank you!

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END